Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam F · Form 2: Advanced reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. A robot moves 6 m east, 18 m north, then 4 m west. What is the angle of its displacement north of east?

(A) 167.32 °

(B) 41.83 °

(C) 83.66 °

(D) -83.66 °

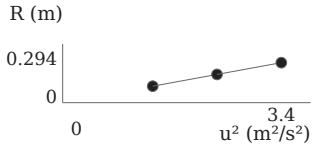
2. Launch at 30° and land at launch height, without drag. Use g=10 m/s². Use 60° while preserving speed and equal launch and landing heights. Which X in the modified apparatus yields Y=0.2078?

(A) 2.49

(B) 2.4

(C) 2.43

(D) 2.34



3. A ball is launched at 15 m/s, 80° above horizontal, and lands at its launch height. Ignore drag. Find the other launch angle producing the same range at this speed. Use g = 9.8 m/s².

(A) 10 °

(B) -10 °

(C) 20 °

(D) 5 °

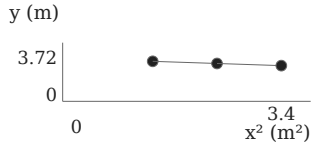
4. A ball leaves a 4 m ledge horizontally at 5 m/s. Use g=10 m/s² and consider airborne frames only. x² (m²) -> y (m): (1.4, 3.72); (2.4, 3.52); (3.4, 3.32). Both sensors multiply their axes by the same unknown positive gain k. What is the uncorrected slope?

(A) -0.2

(B) -0.28

(C) -0.04

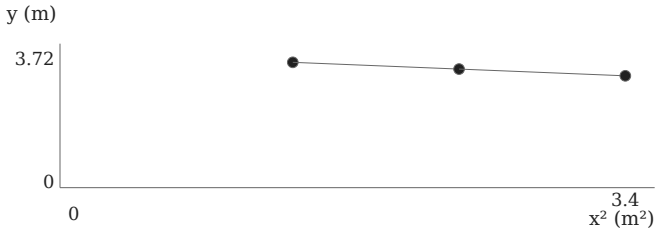
(D) -0.12



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A ball leaves a 4 m ledge horizontally at 5 m/s. Use g=10 m/s² and consider airborne frames only. x² (m²) -> y (m): (1.4, 3.72); (2.4, 3.52); (3.4, 3.32).

a. For the airborne video frames setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Double horizontal launch speed at the same height. Predict the new slope and intercept and explain physically what changed. **(3 pt)**

6. In gravity g=10 m/s² vary the upward component of launch velocity and time the ascent. uy (m/s) -> tpeak (s): (1.4, 0.14); (2.4, 0.24); (3.4, 0.34).

a. Build the model for apex timing from physics and predict both slope and intercept. **(2 pt)**

b. Double gravitational acceleration. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. Launch at 30° and land at launch height, without drag. Use g=10 m/s². u² (m²/s²) -> R (m): (1.4, 0.1212); (2.4, 0.2078); (3.4, 0.2944). Build the model for level-ground range from physics and predict both slope and intercept. **(3 pt)**